



Matrices

Key terms

matrix, equality, addition, scalar, multiplication, determinant, cofactor

KSBs: K13, S11, B1, S53

9.1 Matrices

We have seen how numbers can be stored and manipulated in vectors. Matrices are an extension of this to larger grids of entries.

Definition 9.1 - Matrix

A matrix is a *rectangular array of entries*, which can be numbers, mathematical expressions or symbols.

Matrices are arranged in *rows and columns* and enclosed by round $()$ or square $[]$ brackets. For example:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \begin{bmatrix} 7 & 8 \\ 9 & 10 \end{bmatrix}$$

This matrix contains *6 entries* arranged in *2 horizontal rows* and *3 vertical columns*.

The **dimension** or size of a matrix is written as $m \times n$ where m and n are the number of *rows and columns* of the matrix respectively. We might say it is an "*n by n matrix*".

If $m = n$, so the number of rows and columns are *equal*, we say that the matrix is a *square matrix*.

$$\begin{array}{c} \leftarrow \text{columns} \rightarrow \\ \begin{array}{c} \uparrow \\ \text{rows} \\ \downarrow \end{array} \begin{pmatrix} \square & \square & \dots & \square \\ \square & \square & \dots & \square \\ \vdots & \vdots & \ddots & \vdots \\ \square & \square & \dots & \square \end{pmatrix} \end{array}$$

Matrices are typically given names that are *single uppercase characters, e.g. A* and the entries of a matrix are labelled with the *corresponding lowercase character, e.g. a*.

An individual entry would be denoted a_{ij} where i is the *row number* reading from top to bottom and j is the *column number* reading from left to right.

Example 9.1. If A is an $m \times n$ matrix, then

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

The entry in the *third row, second column* would be a_{32} .

9.2 Operations on Matrices

Definition 9.2 - Matrix Equality

We say that an $m \times n$ matrix A and a $p \times q$ matrix B are **equal** and write $A=B$ if and only if **both** of the following conditions are satisfied:

- they have the *same dimensions, i.e. $m=p$ and $n=q$.*
- for all $1 \leq i \leq m$ and $1 \leq j \leq n$, *$a_{ij} = b_{ij}$.*

Definition 9.3 - Matrix Addition and Subtraction

Let A and B be two $m \times n$ matrices. The sum of two $m \times n$ matrices A and B is an $m \times n$ matrix $A+B$ defined by:

$$[A+B]_{ij} = a_{ij} + b_{ij}.$$

where $1 \leq i \leq m$ and $1 \leq j \leq n$.

Similarly, we can define the difference of two matrices using matrix subtraction:

$$[A-B]_{ij} = a_{ij} - b_{ij}.$$

where $1 \leq i \leq m$ and $1 \leq j \leq n$.

Example 9.2. The sum of two 2×2 matrices is given by matrix addition:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \end{pmatrix}$$

Similarly, the difference is:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}-b_{11} & a_{12}-b_{12} \\ a_{21}-b_{21} & a_{22}-b_{22} \end{pmatrix}$$

Note

The sum and difference of two matrices of different sizes is *not defined*.

Exercise 9.1. Evaluate the following:

$$i) \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$ii) \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 7 \\ -11 \\ -13 \end{pmatrix}$$

$$iii) \begin{pmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \end{pmatrix} + \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

For all $m \times n$ matrices A, B and C , the following conditions are satisfied:

- $A + B = B + A$
- $A + (B + C) = (A + B) + C$.

Definition 9.4 - Scalar multiplication

Let A be a matrix and k be a scalar. The product of the scalar and the matrix, kA , can be found using **scalar multiplication**, defined as:

$$[kA]_{ij} = k \cdot a_{ij}$$

i.e., we multiply each element in the matrix by the scalar.

$$k \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} ka_{11} & \cdots & ka_{1n} \\ \vdots & \ddots & \vdots \\ ka_{m1} & \cdots & ka_{mn} \end{pmatrix}.$$

Exercise 9.2. Evaluate the following:

$$(i) 2 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$(ii) \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 3 & 2 \\ 4 & -2 \end{pmatrix}$$

$$(iii) a \begin{pmatrix} 1 & 6 & 4 \\ 0 & 3 & -1 \end{pmatrix}$$

$$(iv) 101 \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} - 99 \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

Let A and B be two $m \times n$ matrices, and k and ℓ be two scalars. Then

- $kA = AK$
- $k(A+B) = kA + kB$
- $(k+\ell)A = kA + \ell A$
- $k(\ell A) = (k\ell)A = \ell(kA)$
- $-1 \times A = -A$

Definition 9.5 - Matrix Transpose

The **transpose** of an $m \times n$ matrix A is an $n \times m$ matrix denoted by A^T formed by switching the rows and columns of A , i.e.,

$$[A^T]_{ij} = a_{ji}$$

Transposing a matrix switches the rows and columns around so that row i becomes column i and column j becomes row j . i.e.,

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}^T = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{mn} \end{pmatrix}$$

Example 9.3.

$$\begin{matrix} 2 \times 3 \\ \begin{pmatrix} 2 & 0 & 3 \\ 4 & 1 & 5 \end{pmatrix}^T \end{matrix} = \begin{matrix} \begin{pmatrix} 2 & 4 \\ 0 & 1 \\ 3 & 5 \end{pmatrix} \\ 3 \times 2 \end{matrix}$$

Note that what was previously the **first row of the matrix** is now the **first column** and so on. The number of rows and columns in the matrix have been **swapped**.

Exercise 9.3. Evaluate the following:

(i) $\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}^T$

(ii) $\begin{pmatrix} 1 & 0 & -2 \\ 3 & -4 & 1 \end{pmatrix}^T$

(iii) $\begin{pmatrix} 2 & 3 & 5 \end{pmatrix}^T$

Let A and B be two square $n \times n$ matrices and k a scalar. Then

- $(A^T)^T = A$
- $(A+B)^T = A^T + B^T$
- $(kA)^T = k(A^T)$

9.3 Matrix Multiplication

We have seen that we can perform the arithmetic operations of addition and subtraction on matrices, and multiply a matrix by a scalar. Here we will look at *multiplication of two matrices*.

Definition 9.6 - Matrix Multiplication

Let A be an $m \times n$ matrix and B a $p \times q$ matrix, the product AB is defined as

$$[AB]_{ij} = \sum_k a_{ik} b_{kj}$$

The product AB is only defined if *the number of columns in the first matrix* is the same as *the number of rows in the second matrix* i.e., $n=p$, and the resulting matrix has *the same number of rows* as the first matrix and *the same number of columns* as the second matrix, i.e., a $m \times q$ matrix.

Example 9.4. The following pairs of matrices can be multiplied together:

$$(2 \times 2)(2 \times 2) \quad (4 \times 3)(3 \times 2) \quad (3 \times 3)(3 \times 5)$$

These pairs cannot:

$$(2 \times 2)(4 \times 3) \quad (4 \times 5)(3 \times 1) \quad (1 \times 3)(2 \times 2)$$

To check whether matrix multiplication is defined for a given pair of matrices, one easy check is to *write the dimensions of the matrices* next to each other, e.g. underneath the matrices, and if the two *inside numbers* are the same, the matrix multiplication can be performed.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} \quad \begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix} \begin{pmatrix} g & h & i & j \\ k & l & m & n \end{pmatrix}$$

$2 \times 2 \quad 2 \times 2 \qquad 3 \times 2 \quad 2 \times 4$

Furthermore, the dimensions of the product of these two matrices is given by the *two outside numbers* - in the examples above, we'd get a 2×2 matrix and a 3×4 matrix.

9.3.1 Multiplying two matrices

The technique used to multiply two matrices together requires us to work across the *horizontal rows* of the first matrix (the i index) and down the *vertical columns* of the second matrix (the j index).

Example 9.5. Consider the product of the two 2×2 matrices A and B given below:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

The first thing we need to do is check whether matrix multiplication is defined for these matrices. They are 2×2 and 2×2 so this works.

Using the rule given in the matrix multiplication equation to calculate $[AB]_{11}$ (the $(1,1)$ th entry in the product AB), we use row $i=1$ and column $j=1$.

We move through the elements from row 1 of A and column 1 of B multiplying the corresponding values and summing the result:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1 \times 5 + 2 \times 7 & \square \\ \square & \square \end{pmatrix} = \begin{pmatrix} 5 + 14 & \square \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & \square \\ \square & \square \end{pmatrix}$$

We can write this as:

$$[AB]_{11} = a_{11}b_{11} + a_{12}b_{21} = 19$$

Carrying on across the top row of our product matrix AB : next we need to calculate the value of $[AB]_{12}$

Now our row number is $i=1$ and column number is $j=2$ so we multiply the values from row 1 of A and column 2 of B and sum the result.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 1 \times 6 + 2 \times 8 \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & 6 + 16 \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ \square & \square \end{pmatrix}$$

Or in other words,

$$[AB]_{12} = a_{11}b_{12} + a_{12}b_{22}$$

We can continue this process to fill in the rest of the matrix.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 3 \times 5 + 4 \times 7 & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 15 + 28 & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & \square \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 3 \times 6 + 4 \times 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 18 + 32 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}.$$

Multiplying together larger matrices simply involves adding together the products of *all the entries in each row and column*, and repeating this for each entry in the product matrix.

This is why we require the number of columns in the first matrix to be the same as the rows in the second: we need to multiply together *one from each* and add the products.

Exercise 9.4. Given the matrices

$$A = \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 3 & -2 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

calculate the following (where possible):

(i) AB ;

(ii) BC ;

(iii) CD ;

(iv) CC^T

The following properties hold for matrix multiplication:

-
- $AB \neq BA$
 - $A(BC) = (AB)C$
 - $A(B + C) = AB + AC$
 - $(A + B)C = AC + BC$
 - $(AB)^T = B^T A^T$

9.4 Determinants

A **determinant** is a *scalar value* that is calculated using the elements of a *square matrix*. (non-square matrices do not have a determinant). Determinants play a very important role in linear algebra, and tell us about the properties of the matrix. One of their uses is they enable us to determine if a system of linear equations has a *unique solution* (as we will see later).

9.4.1 Calculating the determinant of a 2×2 matrix

Definition 9.7 - Determinant of a 2×2 matrix

The determinant of a square matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, denoted by $\det(A)$ or $|A|$, is

$$\det(A) = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

i.e., the product of the entries on the *main diagonal* minus the product of the *other two entries*.

Note

The formula for the determinant, *ad-bc* will be a useful thing to remember. Come up with your own way of remembering it!

Example 9.6. The determinant of the matrix $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ is given by:

$$\det(A) = \begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} = (1 \times 3) - (2 \times 0) = 3 - 0 = 3$$

Exercise 9.5. Calculate the following determinants:

$$(i) \begin{vmatrix} 5 & 2 \\ 3 & 4 \end{vmatrix}$$

$$(ii) \det \begin{pmatrix} a & b \\ ka & kb \end{pmatrix}$$

$$(iii) \begin{vmatrix} 2-\lambda & 3 \\ 5 & 6-\lambda \end{vmatrix}$$

9.4.2 Determinant of $n \times n$ matrices

Computing the determinant of a 2×2 matrix is relatively straightforward, but computing the determinant of a matrix *larger than 2×2* gets more complicated.

Essentially, we split the larger matrix up into *multiple 2×2 matrices* so we can use the equation for a 2×2 determinant repeatedly to obtain the overall determinant. This is done using **minors** and **cofactors**.

Definition 9.8 - Minor

The **minor** of an element of an $n \times n$ square matrix is denoted by M_{ij} and is the determinant of the $(n-1) \times (n-1)$ square matrix that is formed by *removing row i and column j from the matrix.*

Example 9.7. Given the matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

then the minor M_{21} is the determinant of the matrix A with *row 2 and column 1 removed.*

$$M_{21} = \begin{vmatrix} \square & a_{12} & a_{13} \\ \square & \square & \square \\ \square & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = a_{12}a_{33} - a_{13}a_{32}$$

Exercise 9.6. Given the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

calculate:

(i) M_{11} ; $\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix}$

(ii) M_{12} ;

(iii) M_{13}

9.4.3 Cofactors

Definition 9.9 - Cofactor

The **cofactor** of an element of a square matrix is denoted by C_{ij} and is defined as

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The $(-1)^{i+j}$ term in this equation is *positive* when ij is *even* and *negative* when ij is *odd*, which results in the following pattern of signs:

$$\begin{pmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Example 9.8. Given the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Then

$$\begin{aligned} C_{21} &= (-1)^{2+1} M_{21} \\ &= (-1)^3 M_{21} = -M_{21} \\ &= -\det \begin{pmatrix} 2 & 3 \\ 8 & 9 \end{pmatrix} = -(18 - 24) = -(-6) = 6 \end{aligned}$$

Now we have the terminology of minors and cofactors, we can use them to calculate determinants. For a larger matrix, we choose a single row or column of the matrix, and work our way along it.

Definition 9.10 - Determinant of an $n \times n$ matrix

The determinant of an $n \times n$ matrix A is defined by

$$\det(A) = \sum_i a_{ik} C_{ik} = \sum_j a_{kj} C_{kj}$$

for some fixed value k in the range $1 \leq k \leq n$ which represents a single row or column of A .

This equation allows us to express the determinant of an $n \times n$ matrix in terms of *the determinants of $(n-1) \times (n-1)$ matrices*. We can then apply the formula again to the sub-matrices. Continuing in this fashion we will eventually just be calculating determinants of 2×2 matrices, which we know how to do.

Example 9.9. To calculate the determinant of the matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

we can *expand across row 1* - this means we are using $k=1$ in the ^{second} summation. *(this)*

$$\begin{aligned} \det(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= (-1)^{1+1} \begin{vmatrix} e & f \\ h & i \end{vmatrix} a + (-1)^{1+2} \begin{vmatrix} d & f \\ g & i \end{vmatrix} b + (-1)^{1+3} \begin{vmatrix} d & e \\ g & h \end{vmatrix} c \end{aligned}$$

This is just the *alternating sum* of the determinants of *three 2×2 submatrices* of A , each multiplied by the entry defining which rows and columns are not used in that matrix.

We could also have chosen to expand along *column 2* using $k=2$ and the *other* summation:

$$\begin{aligned} \det(A) &= (-1)^{1+2} b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + (-1)^{2+2} e \begin{vmatrix} a & c \\ g & i \end{vmatrix} + (-1)^{3+2} h \begin{vmatrix} a & c \\ d & f \end{vmatrix} \\ &= -b(di - fg) + e(ai - cg) - h(af - cd) \end{aligned}$$

In our first example, the final line can be multiplied out to give $aei - afh - bdi + bfg + cdh - ceg$, and this is the same (after some rearrangement) as the result when we expand along column 2.

Note

It does not matter which row or column we expand along to compute the determinant - we will always get the same answer. It is usually preferable to expand along the row or column with the most zero elements or the smallest numbers to simplify the calculations.

For larger matrices we have to apply the equation recursively until we get to 2×2 determinants where we can use our 2×2 determinant equation.

Exercise 9.7. (a) Calculate the determinant of the matrix

$$\begin{pmatrix} 1 & 0 & 4 \\ 2 & 5 & 6 \\ 4 & 5 & 2 \end{pmatrix},$$

by expanding along:

(i) row 1;

$$\begin{vmatrix} 1 & 0 & 4 \\ 2 & 5 & 6 \\ 4 & 5 & 2 \end{vmatrix} = 1 \begin{vmatrix} 5 & 6 \\ 5 & 2 \end{vmatrix} - 0 \begin{vmatrix} 2 & 6 \\ 4 & 2 \end{vmatrix} + 4 \begin{vmatrix} 2 & 5 \\ 4 & 5 \end{vmatrix} = (5 \times 2 - 5 \times 6) + 4(2 \times 5 - 4 \times 5) \\ = (10 - 30) + 4(10 - 20) = -60$$

(ii) column 3

(b) Calculate the determinant of the 4×4 matrix

$$\begin{pmatrix} 1 & -1 & 4 & 3 \\ 2 & 0 & 5 & -3 \\ 1 & 2 & 4 & 5 \\ 2 & 0 & -2 & 4 \end{pmatrix}$$

Determinants have the following properties:

- $\det(AB) = \det(A)\det(B)$
- $\det(A) = \det(A^T)$
- If a matrix M has a row or column with all zero elements then $\det(M) = 0$
- Interchanging any two rows of a matrix changes the sign of $\det(M)$